



# Deploy Backend with Kubernetes



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```
[ec2-user@ip-172-31-10-27 manifests]$ kubectl apply -f flask-deployment.yaml
kubectl apply -f flask-service.yaml
deployment.apps/nextwork-flask-backend created
service/nextwork-flask-backend created
```



# Introducing Today's Project!

In this project, I will deploy the backend of an application with Kubernetes. This is because this is an important skill that DevOps engineers need as this is what happens in the real world.

## Tools and concepts

I used Kubernetes, ECR, kubectl, Docker, eksctl, GitHub to deploy the backend of an application. Key concepts include using manifest files, building container images and pushing them to a container registry.

## Project reflection

This project took me approximately 1 hour and 20 minutes. The most challenging part was creating the manifest files. My favourite part was seeing that it had been deployed in the terminal where it said the resources have been created.



# Project Set Up

## Kubernetes cluster

To set up today's project, I launched a Kubernetes cluster. The cluster's role in this deployment is to be the control centre for using Kubernetes and orchestrating all the containers that will be running the application I want to deploy

## Backend code

I retrieved backend code by cloning a GitHub repository so I can have a local copy of that code. Pulling code is essential to this deployment because I need to access it to build a container image of the application so that Kubernetes will deploy it.

## Container image

Once I cloned the backend code, I built a container image because needs the application to be packaged in an image to deploy it. Without an image, it would be difficult for Kubernetes to deploy an application consistently across a cluster.

I also pushed the container image to a container registry, which is a storage space for container images in the cloud. ECR facilitates scaling for my deployment because Kubernetes can easily use whichever image is tagged latest for each container it creates.



# Manifest files

Kubernetes manifests are the instructions/manuals on how we would like Kubernetes to deploy, expose and run a containerised application. Manifests are helpful because it allows Kubernetes to repeat a deployment using the same instructions.

A Deployment manifest manages the creation/setting up of a deployment resource that is responsible for creating containers/rolling out updates across my EKS cluster. The container image URL in my Deployment manifest tells Kubernetes where my image is.

A Service resource exposes a deployed application to users/traffic. My Service manifest sets up Service resource that exposes my backend application at port 8080 within the EKS cluster.

```
--
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nextwork-flask-backend
  namespace: default
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nextwork-flask-backend
  template:
    metadata:
      labels:
        app: nextwork-flask-backend
    spec:
      containers:
        - name: nextwork-flask-backend
          image: 840032438341.dkr.ecr.eu-west-2.amazonaws.com/nextwork-flask-backend:latest
          ports:
            - containerPort: 8080
~
~
```



# Backend Deployment!

To deploy my backend application, I applied the manifest files that I created. Applying a manifest file means giving Kubernetes the templates for creating the Deployment and Services resources to deploy and expose my application.

## kubectl

kubectl is the command line tool for managing resources inside an EKS cluster. I need this tool to deploy the application by creating the Deployment/Service resources. I can't use eksctl for the job because that tool is for creating the cluster.

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